

CMPT 225-D100 Quiz 5, Summer 2026

Duration	30 minutes for the entire exam.
Coding Questions	For coding questions, use C++-like pseudocode. This means that small coding errors will usually be overlooked. However, it must be clear that the code could easily be converted to real C++.
Aids allowed	Pencil or pen, and eraser. No notes, no papers, no books, no computers, no calculators, no cheat sheets, etc.
During the exam	Raise your hand if you would like to speak with a proctor. To be fair to all students, we don't answer questions about the exam. Choose the answer based on your best understanding of the question. Re-reading the question may help.

For each multiple choice question, choose **the 1 best answer**. Each correct answer is worth 1 mark. Unanswered, illegible, or ambiguous answers are worth 0 marks.

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| 1. | i) A tree has no cycles.
ii) A tree is connected. | a) i) and ii) are both true
b) i) and ii) are both false | c) i) is true and ii) is false
d) i) is false and ii) is true |
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Answer: a) A tree is a graph that is both connected and does not have any cycles.

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| 2. | On a simple graph with m edges and n vertices, what is the worst-case running time of depth -first search? | a) $O(m)$
b) $O(n)$ | c) $O(m + n)$
d) $O(m \cdot n)$ |
|----|---|------------------------|------------------------------------|

Answer: c)

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| 3. | On a simple graph with m edges and n vertices, what is the worst-case running time of breadth -first search? | a) $O(m)$
b) $O(n)$ | c) $O(m + n)$
d) $O(m \cdot n)$ |
|----|---|------------------------|------------------------------------|

Answer: c)

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|----|---------------------------------|---------------------------------|
| 4. | How many edges does K_5 have? | a) 5
b) 10
c) 15
d) 20 |
|----|---------------------------------|---------------------------------|

Answer: d) K_n always has $\frac{n(n-1)}{2}$ edges.

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| 5. | i) The discovery edges from running depth-first search on a simple graph form a spanning tree.
ii) The back edges from running depth-first search on a simple graph form a spanning tree. | a) i) and ii) are both true
b) i) and ii) are both false | c) i) is true and ii) is false
d) i) is false and ii) is true |
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Answer: c) The discovery edges form a spanning tree, but the back edges do not (e.g. a tree has no back edges).

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| 6. | If the sum of the degrees of all vertices of a simple graph is 100, how many vertices does it have? | a) 50
b) 100
c) 200 | d) can't tell from given information |
|----|--|---------------------------|--------------------------------------|

Answer: d) We can't determine the number of vertices from the information given.

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| 7. | If the sum of the degrees of all vertices of a simple graph is 100, how many edges does it have? | a) 50
b) 100
c) 200 | d) can't tell from given information |
|----|---|---------------------------|--------------------------------------|

Answer: a) The sum of the degrees of all vertices is twice the number of edges.

8. i) Dijkstra's algorithm can find the shortest path in a weighted graph (with non-negative edge weights) between a given starting vertex and all other vertices.
ii) Dijkstra's algorithm runs in worst case $O(m + n)$ time, where m is the number of edges and n is the number of vertices.

a) i) and ii) are both true c) i) is true and ii) is false
b) i) and ii) are both false d) i) is false and ii) is true

Answer: c) Dijkstra's algorithm can find the shortest path in a weighted graph (with non-negative edge weights) between a given starting vertex and all other vertices. However, its worst case running time is $O((m + n) \log n)$.

9. i) Depth-first search can be used to test if a graph is connected.
ii) Breadth-first search can be used to test if a graph is connected.

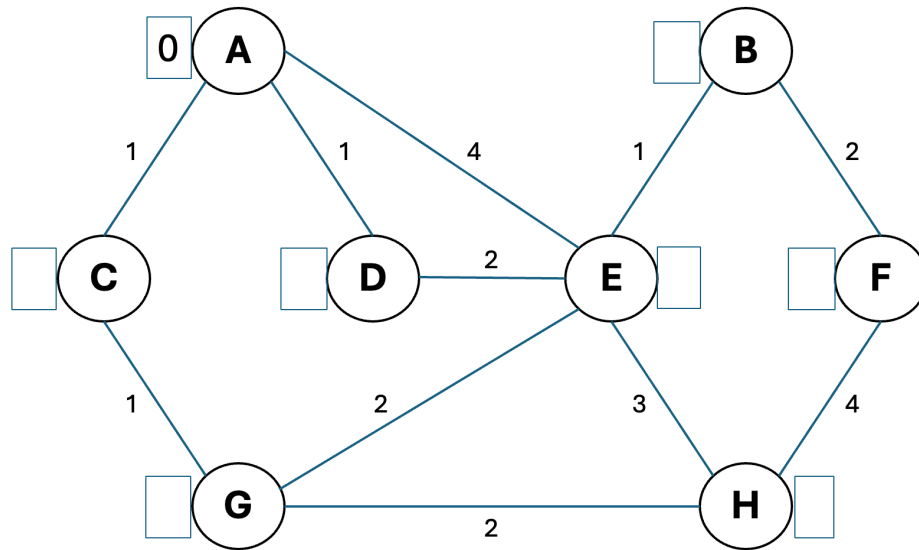
a) i) and ii) are both true c) i) is true and ii) is false
b) i) and ii) are both false d) i) is false and ii) is true

Answer: a) Both depth-first search can be used to test if a graph is connected.

10. What's the worst-case running time of Kruskal's algorithm on a graph with m edges and n vertices?
- a) $O(m + n)$ b) $O(mn)$ c) $O(\log(m + n))$ d) none of the others

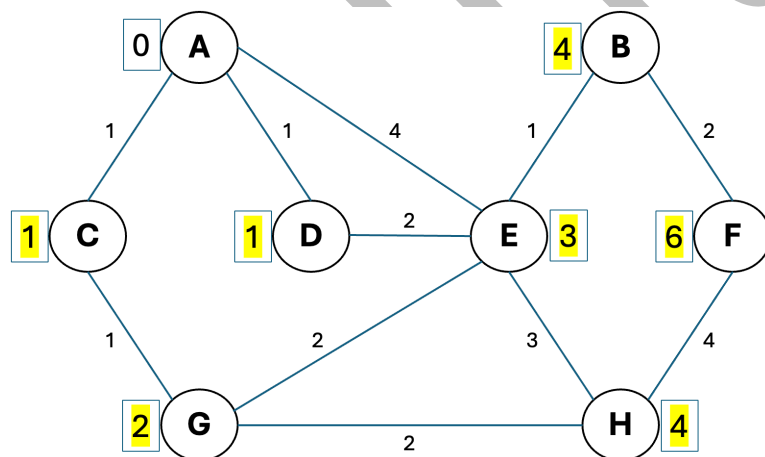
Answer d): Kruskal's algorithm runs in worst-case $O((m + n) \log n)$ time.

11. In each of the boxes below, write the length of the shortest path from **A** to that node. Use **Dijkstra's algorithm**, and in the table below list the vertices in the order they are removed from the queue in the algorithm. **Important:** in case of ties, **always choose the vertex that comes first alphabetically**.

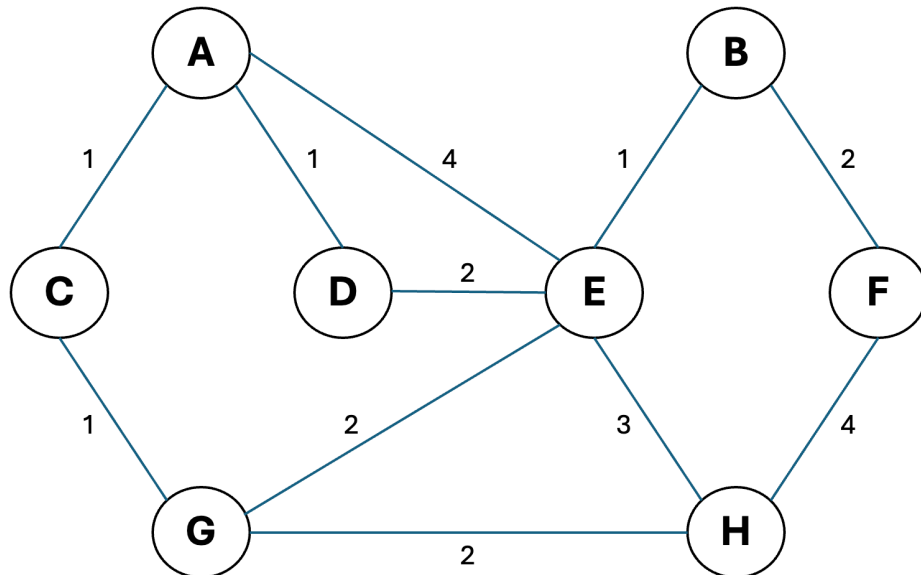


Order that vertices are removed from the queue:

1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
A	C	D	G	E	B	H	F

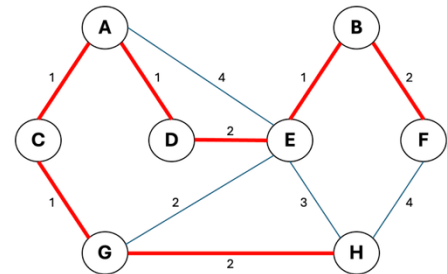


12. On the copy of the graph given underneath, draw the minimum spanning tree that **Kruskal's algorithm** finds for this graph (include the edge weights, but **only the spanning tree edges**). **Important:** name all edges as the alphabetical ordering of their end vertices. For example, an edge with end vertices **T** and **M** is named **MT**. Whenever there is a tie, always choose the edge whose name (as described) comes first alphabetically.



A

B



C

D

E

F

G

H